

CONCORDIA BLOSSER, KS, USA

WMO: 724580

Lat: 39.551N

Lon: 97.651W

Elev: 448

StdP: 96.06

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13984

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-14.2	-21.6	0.6	-15.5	-19.2	0.7	-13.0	12.7	-0.3	11.7	1.3	4.7	350	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	36.9	23.3	35.0	23.2	33.4	22.9	25.5	32.4	24.7	31.7	24.0	30.9	6.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.4	29.0	22.7	18.4	28.2	21.9	17.5	27.5	80.8	32.5	77.3	31.6	74.3	30.9	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.0	10.9	9.5	DB	-20.5	39.5	2.1	1.6	-22.1	40.6	-23.3	41.5	-24.5	42.4	-26.1	43.5	
			WB	-20.9	26.8	2.0	1.1	-22.4	27.6	-23.6	28.2	-24.8	28.8	-26.2	29.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-1.7	0.3	6.3	11.8	17.6	23.7	26.2	24.8	20.5	13.4	6.1	-0.3
	DBStd	11.12	6.44	6.92	6.60	5.39	4.68	3.79	3.42	3.39	4.76	5.24	5.60	5.96
	HDD10.0	1321	363	276	149	41	2	0	0	0	1	26	141	321
	HDD18.3	2922	621	503	373	202	72	5	1	1	32	168	368	577
	CDD10.0	2219	1	6	36	97	238	410	502	458	316	131	23	1
	CDD18.3	777	0	0	1	8	49	165	244	200	97	14	0	0
	CDH23.3	8376	0	1	22	104	501	1771	2798	2064	969	142	4	0
	CDH26.7	3851	0	0	3	24	171	812	1437	969	397	37	1	0
Wind	WSAvg	4.7	4.6	4.9	5.3	5.6	4.9	4.8	4.3	3.9	4.5	4.7	4.8	4.5
Precipitation	PrecAvg	732	16	21	48	62	107	109	96	90	72	51	28	24
	PrecMax	1130	47	64	211	152	247	359	425	272	216	134	124	91
	PrecMin	369	0	0	1	9	7	25	3	9	5	0	0	0
	PrecStd	179	11	17	41	31	58	59	77	51	47	35	27	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	21.3	25.9	29.7	33.9	37.2	39.7	38.7	35.9	30.8	23.4	17.0
		MCWB	8.5	10.9	14.7	17.3	20.6	23.3	23.3	22.7	21.6	19.4	14.1	11.1
	2%	DB	12.7	17.0	22.7	26.2	30.8	35.0	37.3	36.2	33.2	26.9	20.1	13.4
		MCWB	6.3	9.5	13.6	16.3	20.1	23.0	23.6	23.5	22.2	17.3	12.7	8.4
	5%	DB	10.0	13.6	19.8	23.6	28.4	33.3	35.4	33.9	31.1	24.4	17.4	10.7
		MCWB	5.0	7.9	12.4	15.3	19.6	22.5	23.7	23.5	21.2	16.4	11.5	6.2
	10%	DB	7.4	10.5	16.9	21.2	26.1	31.4	33.5	31.8	28.8	22.1	14.9	8.3
		MCWB	3.6	5.8	11.0	14.2	18.8	22.1	23.5	23.0	20.2	15.4	10.0	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	13.3	17.7	19.7	23.2	25.7	26.8	26.6	24.3	21.7	15.8	13.0
		MCDB	14.3	17.4	22.9	25.8	29.8	32.4	33.6	32.9	31.1	27.2	19.8	14.6
	2%	WB	7.0	10.5	15.7	18.0	21.9	24.6	25.7	25.3	23.2	19.7	14.2	9.0
		MCDB	11.7	16.0	19.6	23.3	28.0	32.1	32.8	31.9	30.0	24.0	17.8	12.2
	5%	WB	5.3	8.1	13.6	16.5	20.8	23.7	24.9	24.5	22.3	17.9	12.2	6.5
		MCDB	9.7	13.4	18.8	22.4	26.4	30.9	32.1	31.1	28.9	22.3	16.6	10.4
	10%	WB	3.7	5.9	11.2	14.9	19.7	22.8	24.1	23.6	21.4	16.2	10.3	4.8
		MCDB	7.3	10.3	16.4	20.5	24.8	29.6	31.3	29.9	27.2	20.9	14.8	8.2
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.1	12.5	12.4	11.8	12.2	11.9	11.8	12.5	12.5	11.4	10.0
		MCDBR	14.8	16.0	16.4	15.9	14.7	14.1	14.1	14.1	14.0	15.3	14.6	14.1
		MCWBR	9.5	9.9	9.2	8.8	6.7	5.3	4.8	5.0	5.4	7.8	8.6	9.4
	5% WB	MCDBR	14.0	15.0	14.1	14.0	12.8	12.8	12.3	12.1	12.4	12.3	13.1	13.3
		MCWBR	9.4	9.8	8.8	8.8	6.7	5.7	5.3	5.5	5.5	7.4	8.7	9.4
Clear-Sky Solar Irradiance	taub	0.282	0.292	0.324	0.358	0.386	0.399	0.415	0.412	0.373	0.331	0.287	0.275	
	taud	2.491	2.466	2.411	2.345	2.344	2.348	2.348	2.323	2.330	2.385	2.483	2.530	2.506
	Ebn at Noon	899	937	933	917	893	878	860	854	869	877	886	880	
	Edh at Noon	76	90	106	121	124	124	126	122	108	88	73	70	
All-Sky Solar Radiation	RadAvg	2.33	3.12	4.16	5.23	6.01	6.81	6.71	5.93	4.88	3.54	2.58	1.99	
	RadStd	0.19	0.27	0.31	0.29	0.41	0.33	0.41	0.32	0.28	0.35	0.20	0.19	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page